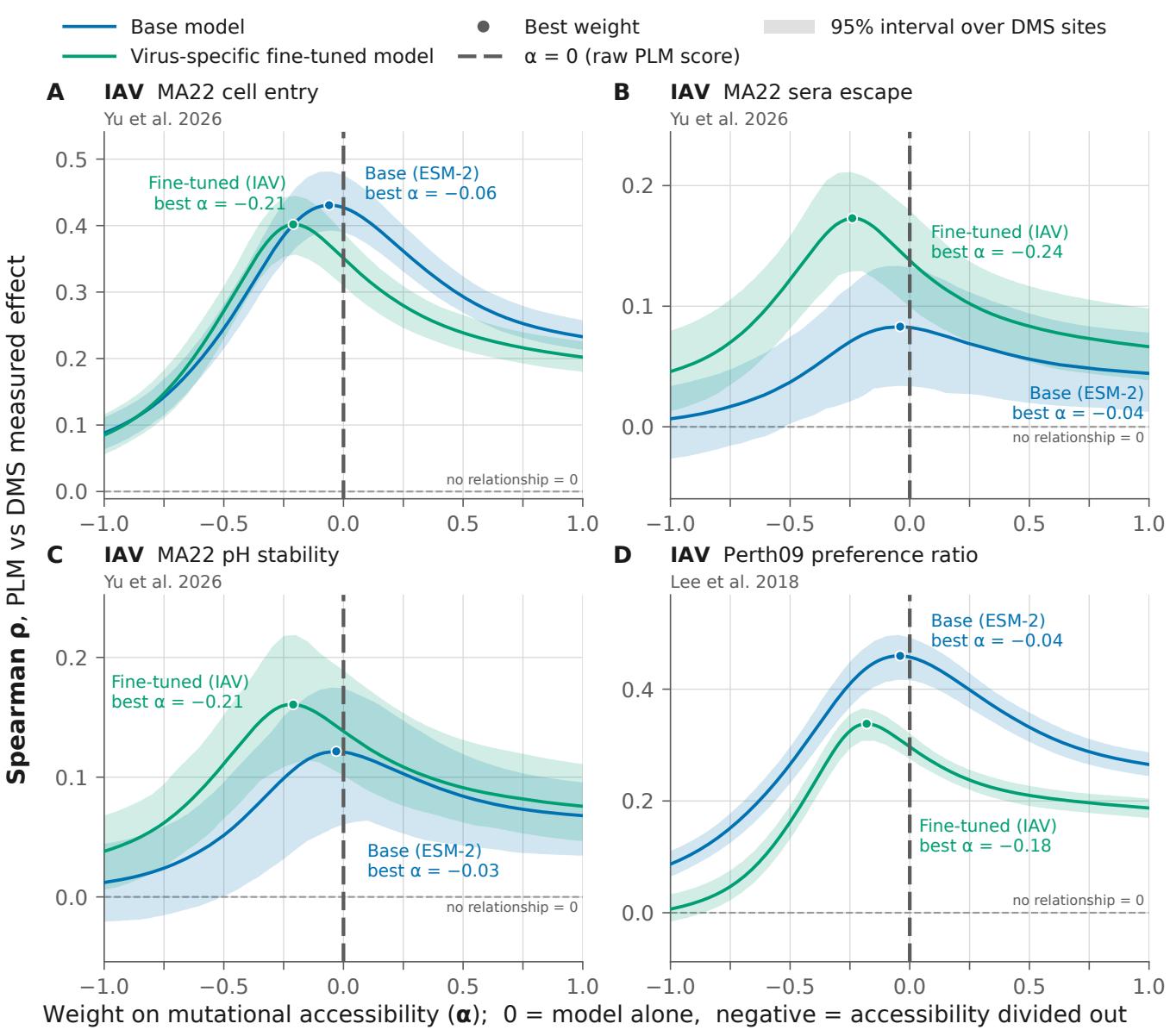

Base model
Virus-specific fine-tuned model
Best weight
α = 0 (raw PLM score)
95% interval over DMS sites
A IAV MA22 cell entry
Yu et al. 2026
Fine-tuned (IAV)
best α = −0.21
Base (ESM-2)
best α = −0.06
no relationship = 0
B IAV MA22 sera escape
Yu et al. 2026
Fine-tuned (IAV)
best α = −0.24
Base (ESM-2)
best α = −0.04
no relationship = 0
C IAV MA22 pH stability
Yu et al. 2026
Fine-tuned (IAV)
best α = −0.21
Base (ESM-2)
best α = −0.03
no relationship = 0
D IAV Perth09 preference ratio
Lee et al. 2018
Base (ESM-2)
best α = −0.04
Fine-tuned (IAV)
best α = −0.18
no relationship = 0
Spearman ρ, PLM vs DMS measured effect
Weight on mutational accessibility (α); 0 = model alone, negative = accessibility divided out